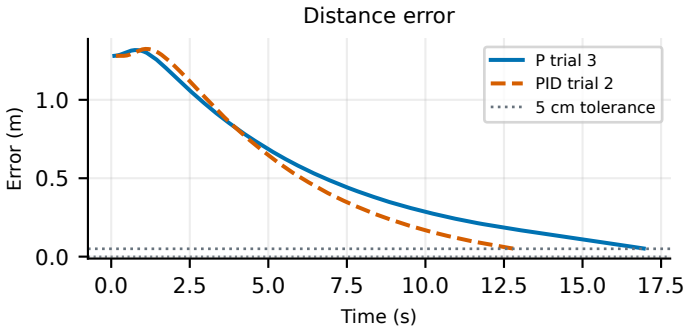
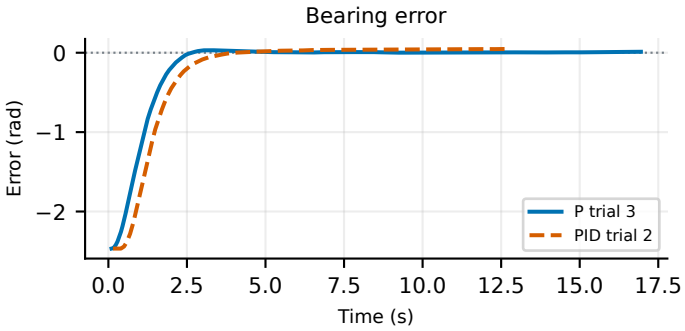
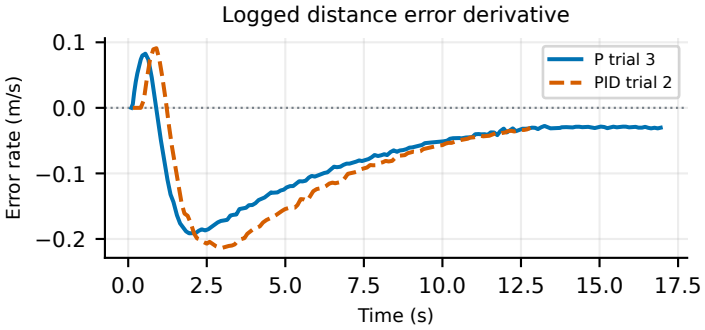
1. Experimental plots

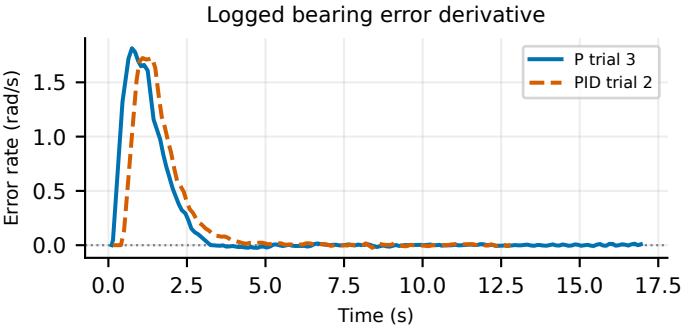
Point control: selected P and PID runs

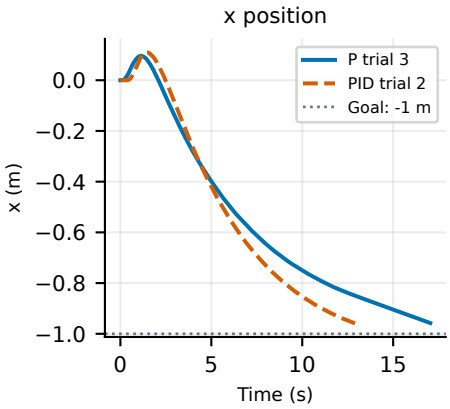
Goal (−1.0, −0.8) m; 5 cm stopping tolerance. Curves use the recorded run labels.

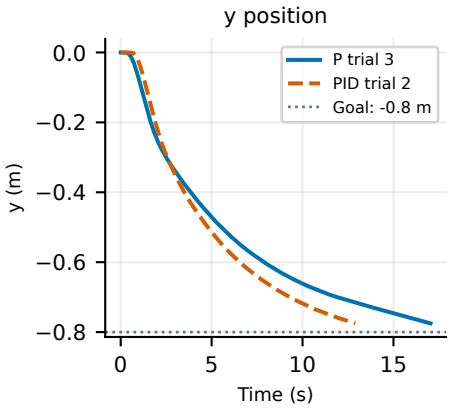


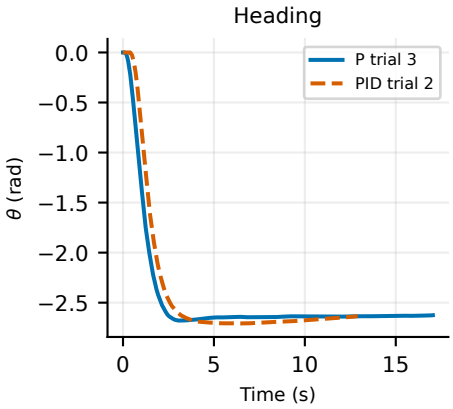


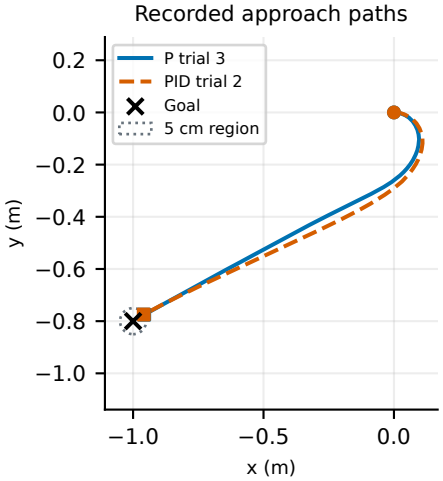


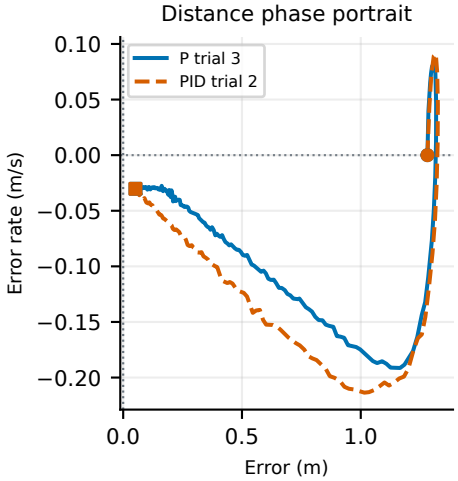


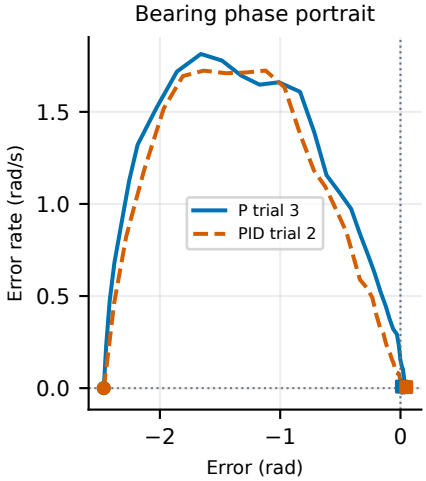






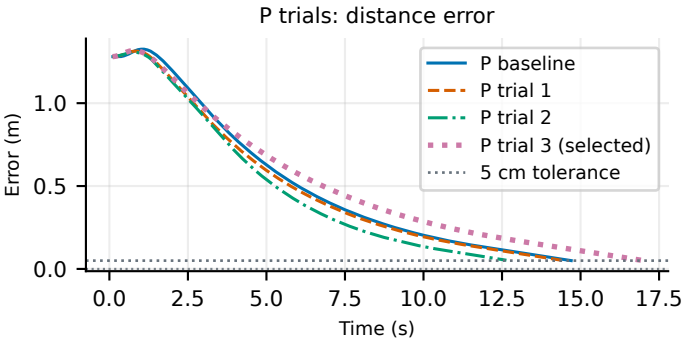


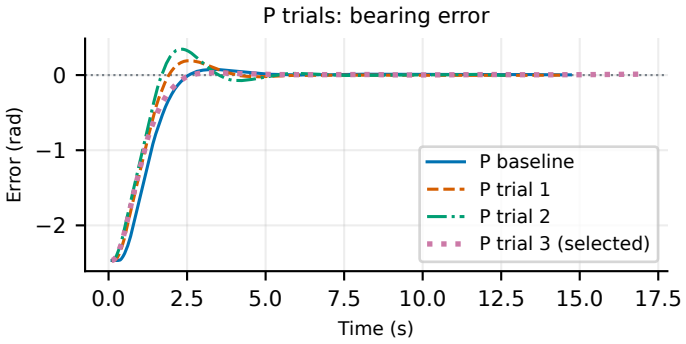


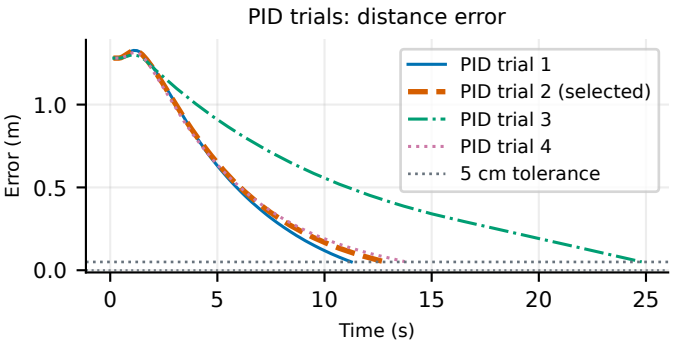


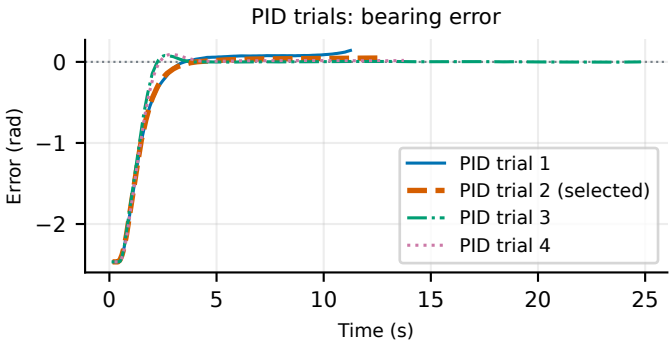
Selected P trial 3 and PID trial 2 have the smallest angular-error overshoot within their respective Part 5 trial groups. Dotted lines mark zero error, position targets or the 5 cm region. Circles and squares mark first and last samples in path/phase plots. The derivative is the logged controller estimate. Each run uses its first pose timestamp as t = 0. Logs end at the position stopping threshold; final heading is not independently controlled.

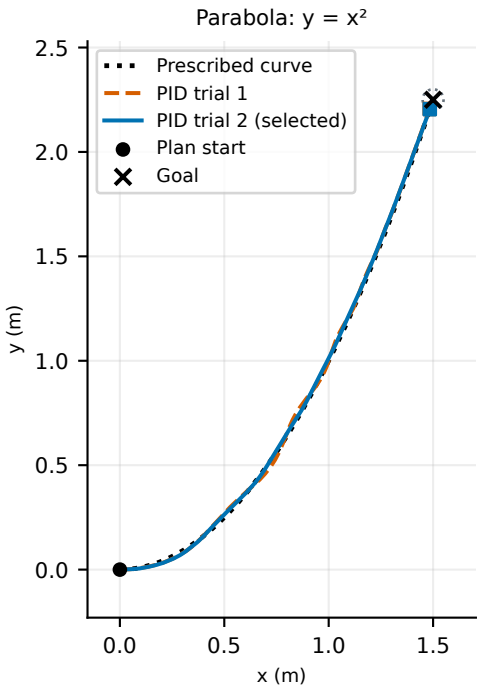
1. Experimental plots — tuning and paths

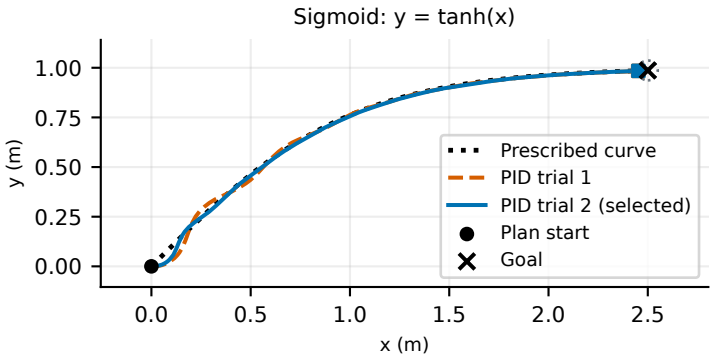












Point tuning: P trial 3 records 1.80° peak angular-error overshoot and arrives in 17.04 s; PID trial 2 records 2.75° and 12.90 s. Other trials expose the speed/overshoot tradeoff. Per-trial gains were not archived, so the plots support response comparisons rather than attribution to a specific gain change.

Trajectory choice: retain PID, which completed both shapes within the 5 cm goal tolerance. Select trial 2 for each shape, prioritising path tracking. Time-weighted RMS distance to the prescribed curve is 0.51 cm for the parabola and 0.65 cm for the sigmoid. Both improve tracking relative to trial 1 but take longer. All in-lab trajectory trials used PID; no in-lab P trajectory comparison is available.